CHAU LAM YU

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RESEARCH INTERESTS

My research integrates geophysical fluid dynamics, high-resolution numerical modeling, and machine learning to understand the multiscale processes governing extreme weather and air-sea interaction. By treating the atmosphere and ocean as a single, integrated dynamical system, my research bridges the gap between extreme weather and oceanic responses to enhance our understanding of storm predictability and the modeling of coastal climate risks.

Dynamics and theory of Tropical Cyclones (TCs)
- Rapid intensification of TC
- TC-shear interaction

Atmosphere-Ocean interaction
- Atmosphere-ocean feedback
- Storm-induced near-inertial wave (NIW) response

Extreme weather and climate risk assessment
- Storm-induced risk of extreme wind, precipitation, and surge

Machine learning and Artificial Intelligence (AI)
- eXplainable AI (XAI) for large-scale AI-weather emulators
- Physics-informed Machine Learning

EMPLOYMENT

Research Scientist — 12/2023 – present
Department of Atmospheric and Environmental Sciences
SUNY at Albany

Postdoctoral Research Associate — 08/2020 – 08/2023
Department of Atmospheric and Environmental Sciences
SUNY at Albany

Research Assistantship — 08/2017 – 06/2020
Department of Meteorology and Atmospheric Science
Penn State University

EDUCATION

Ph.D. in Atmospheric Sciences — 08/2017 – 07/2020
Penn State University
Thesis: The role of asymmetric rainband processes in secondary eyewall formation in tropical cyclones
Advisor: Dr. Anthony Didlake; Co-advisor: Dr. Fuqing Zhang

Master in Atmospheric Sciences — 08/2015 – 07/2017
University of Utah

Master in Mathematics: Numerical modeling of the Atmosphere and Oceans — 08/2012 – 07/2013
University of Reading, UK

Bachelor of Science, double majors in Mathematics and Physics — 08/2009 – 07/2012
University of Hong Kong

PUBLICATIONS

Yu, C.-L. (2026): On the Dynamical and Thermodynamic Constraints of Axisymmetric Tropical Cyclones under Non-Symmetric-Neutrality. *J. Atmos. Sci.* **Accepted**, pre-print available at https://arxiv.org/abs/2502.10474.

Yu, C. L., B. Shrestha and Wang J. (2026). A Novel Bias Correction Scheme for the New York

State Mesonet Microwave Radiometer Temperature Retrieval. *Journal of Atmospheric and Oceanic Technology.* **Minor Revision**.

Yu, C. L., K. Bachmann, R. Torn., A. Brammer and F. D. Marks (2025). Evaluation of the Performance of Probabilistic Precipitation Forecast of PHRaM during 2017-2023 Hurricane Seasons. *Wea. Forecasting, 41, 339–356, https://doi.org/10.1175/WAF-D-25-0123.1.*

Wang J., A. Dai, **C.-L. Yu**, N. Bain, D. J. McGuinnes, and B. Shrestha (2024). Characterizing the Impacts of 2024 Total Solar Eclipse Using New York State Mesonet Data. *Geophy. Res. Letters.*, **51**, e2024GL112684. https://doi.org/10.1029/2024GL112684

Yu, C. L., Tang, B. H., Fovell, R. G. (2024). Tropical Cyclone Boundary Layer Asymmetries in a Tilt-Following Perspective. J. Atmos. Sci., **81**, 1543–1563, https://doi.org/10.1175/JAS-D-23-0178.1.

Yu, C. L., Tang, B. H., & Fovell, R. G. (2023). Diverging Behaviors of Simulated Tropical Cyclones in Moderate Vertical Wind Shear. *J. Atmos. Sci.,* **80**, 2837–2860, https://doi.org/10.1175/JAS-D-23-0048.1.

Yu, C. L., Tang, B. H., & Fovell, R. G. (2023). Tropical cyclone tilt and precession in moderate shear: Precession hiatus in a critical shear regime. *J. Atmos. Sci.* **80**, 909-932. https://doi.org/10.1175/JAS-D-22-0200.1

Yu, C.-L., A. C., Didlake Jr., and F. Zhang (2022), Updraft Maintenance and Axisymmetrization during Secondary Eyewall Formation in a Model Simulation of Hurricane Matthew (2016). *J. Atmos. Sci*. **79**, 1105-1125. https://doi.org/10.1175/JAS-D-21-0103.1

Yu, C.-L., A. C., Didlake Jr., F. Zhang and Robert G. Nystrom (2021), Asymmetric rainband processes leading to secondary eyewall formation in a model simulation of Hurricane Matthew (2016). *J. Atmos. Sci.* **78**, 29-49. https://doi.org/10.1175/JAS-D-20-0061.1

Yu, C.-L., A. C., Didlake Jr., F. Zhang, and J. D., Kepert (2021), Investigating axisymmetric and asymmetric signals of secondary eyewall formation using observations-based modeling of the tropical cyclone boundary layer. Journal of Geophysical Research: Atmospheres. **126**. https://doi.org/10.1029/2020JD034027

Yu, C.-L., and A. C., Didlake Jr. (2019), Impact of stratiform rainband heating on the tropical cyclone wind field in idealized simulations. *J. Atmos. Sci., **76**,* 2443–2462. https://doi.org/10.1175/JAS-D-18-0335.1

Pu, Z., **C.-L. Yu**, V., Tallapragada, J., Jin, and W., McCarty (2019), The Impact of Assimilation of GPM Microwave Imager Clear-Sky Radiance on Numerical Simulations of Hurricanes Joaquin (2015) and Matthew (2016) with the HWRF Model, *Mon. Weather Rev.*, **147**, 175-198. https://doi.org/10.1175/MWR-D-17-0200.1

Yu, C.-L. and M. A. C., Teixeira (2015), Impact of non-hydrostatic effects and trapped lee waves on mountain wave drag in directionally sheared flow. *Quart. J. Roy. Meteorol. Soc.*, **141** (690). pp. 1572-1585. ISSN 1477-870X. https://doi.org/10.1002/qj.2459

Teixeira, M. A. C. and **C.-L. Yu** (2014), The gravity wave momentum flux in hydrostatic flow with directional shear over elliptical mountains. *European Journal of Mechanics & Fluids B: Fluids*, **47**. pp. 16-31. ISSN 0997-7546. https://doi.org/10.1016/j.euromechflu.2014.02.004

Publications In Preparation

Yu, C.-L. (2026): On the hurricane-induced momentum flux convergence and mixing of ocean near-inertia wave. In-prep.

Yu, C. L., Z. Wu. (2026). Does AI-based adjoint sensitivity represent the true adjoint sensitivity of the dynamical system? In-prep.

Other Scholarly Activity

Patent: Bias Correction scheme for Microwave Radiometer temperature retrieval. 2026/001-010.

Contract deliverables: Assessing Climate Exposure and Financial Impacts at the Census Tract Level in New York State. *Department of Financial Services, New York State.*

Grant Funding

NSF Atmospheric Science Cluster (AGS) Program ($727,400), PI: Zheng Wu; Co-PI: Chau Lam Yu, Assessing the Representations of Teleconnections and Understanding Their Roles in Regional Weather Predictability Using AI Weather Prediction Models, 2025.12-2028.11. *Pending*.

Honors and Awards

Research Internal Award: UAlbany Inventor Recognition for First Invention Disclosures, Patent Applications, and Patents Awarded	2025
Student Poster Presentation Award: 34th Conference on Hurricanes and Tropical Meteorology	2020
Scholarship Prize 2013: Most Outstanding Postgraduate Student in the Faculty of Science: University of Reading	2013
Professor C D Beling Memorial Prize in Physics: University of Hong Kong	2012
Dean's Honors List: The University of Hong Kong	2012

Teaching Experience

Guest Lecturer: ATM 317 Dynamic Meteorology II (SUNY Albany)	Spring 2024
Guest Lecturer: ATM 306 Climate Variability and Change (SUNY Albany)	Fall 2023
Guest Lecturer: ATM 316 Dynamic Meteorology I (SUNY Albany)	Fall 2023, 2025
Guest Lecturer: Meteo 452 Tropical Meteorology (PSU)	Fall 2019
Guest Lecturer: Meteo 421 Atmospheric Dynamics (PSU)	Fall 2018
Teaching Assistant: Meteo 421 Atmospheric Dynamics (PSU)	Spring 2018
Teaching Assistant: Meteo 005 Severe Weather and Unusual Weather (PSU)	Fall 2017

Professional Services

Associate Editor:
- Monthly Weather Review (*MWR*)

Reviewer:
- Journal of the Atmospheric Sciences (*JAS*),
- Geophysical Research Letters (*GRL*),
- Monthly Weather Review (*MWR*),
- Quarterly Journal of the Royal Meteorological Society (QJRMS),
- Meteorology and Atmospheric Physics (*MAAP*),
- EGUsphere

Invited talks

Yu, C.-L., (2024). Towards Relaxing the Symmetric Neutrality Assumption of the Potential Intensity Theory of Tropical Cyclones. Nanjing University, China.

Selected Presentations

Yu, C.-L., (2025). Towards Relaxing the Symmetric Neutrality Assumption of the Potential Intensity Theory of Tropical Cyclones. Oral Presentation, *105th AMS Annual Meeting*, *New Orleans, LA.*

Yu, C.-L., Tang, B. H., & Fovell, R. G., (2024). Tilt-following boundary layer asymmetric structure of tropical cyclones in a moderately sheared environment. Oral Presentation, *36th Conference on Hurricanes & Tropical Meteorology, Long Beach, CA.*

Yu, C.-L., A. C., Didlake Jr., F. Zhang, and J. D. Kepert, 2020: Investigating Axisymmetric and Asymmetric Signals of Secondary Eyewall Formation Using Observations-Based Modeling of Tropical Cyclone Boundary Layer. Poster Presentation, *34th Conference on Hurricanes and Tropical Meteorology Virtual Meeting*. (**Student Poster Presentation Award**)

Skills

Modelling Development:
- Non-hydrostatic mountain wave model (Yu and Teixeira 2015)
- Near-inertial ocean wave model (Yu 2026, In-prep)
- Polar Fourier Neural Network (Yu and Wu 2026, In-prep)

Modelling Experience:
- Cloud Model 1 (CM1)
- Weather Research & Forecasting Model (WRF-ARW)
- Hurricane WRF

Languages

- English: Fluent
- Cantonese: Native
- Mandarin: Native